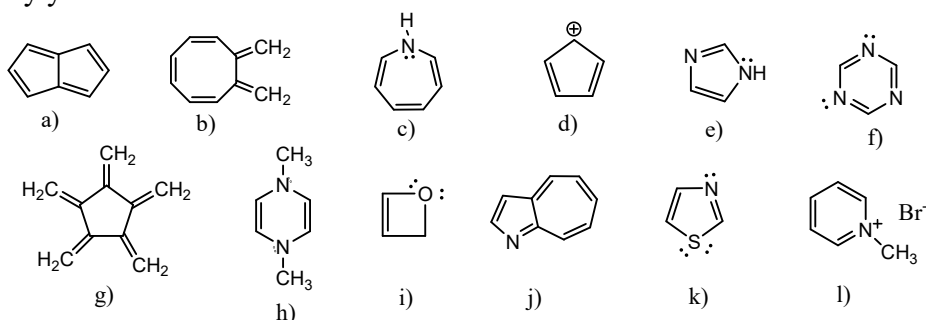


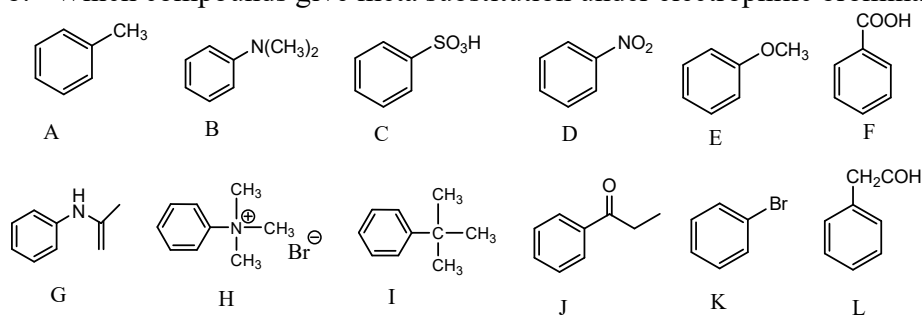
FOC work sheet -IV

1. Classify the following compounds as aromatic or non-aromatic by the application of the Huckel rule. Justify your answer.

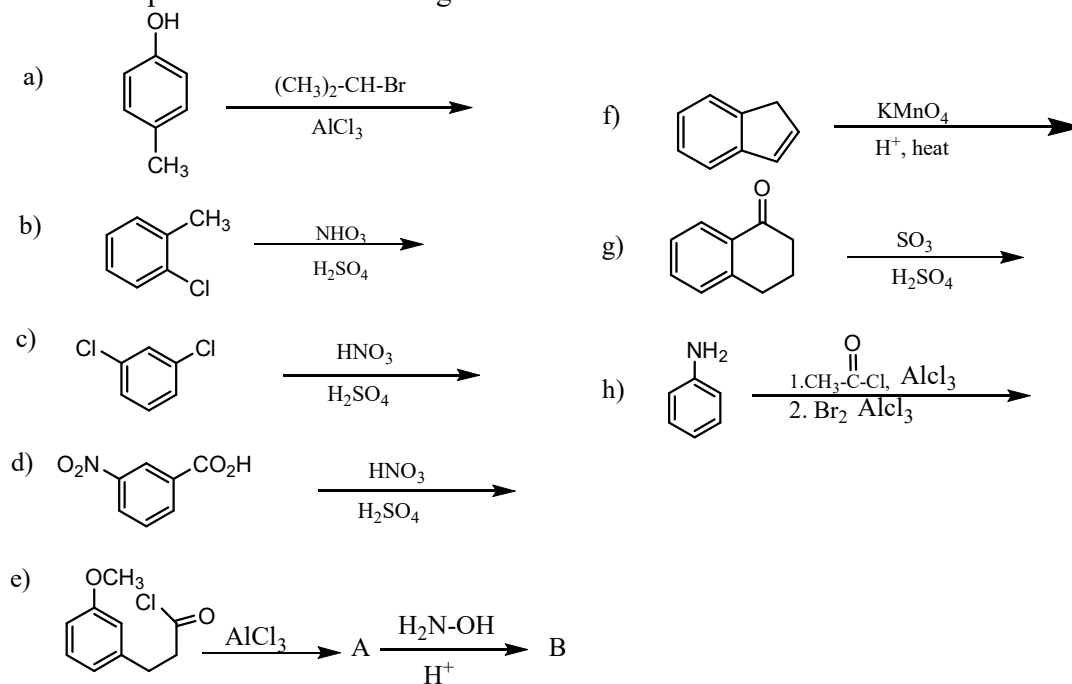


2. Examine the structural formulas shown below and select those that satisfy each of the following conditions.

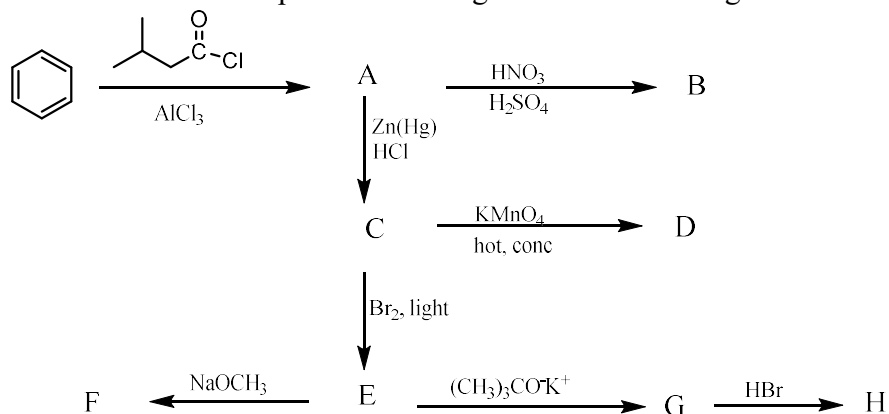
- Which compounds undergo electrophilic nitration more rapidly than benzene?
- Which compounds give meta substitution under electrophilic bromination conditions?



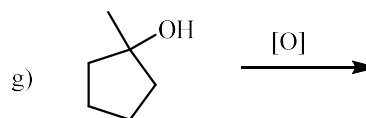
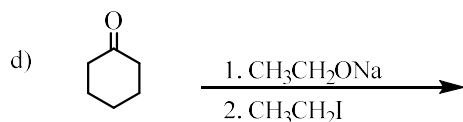
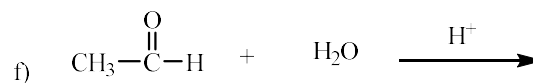
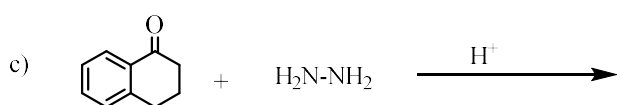
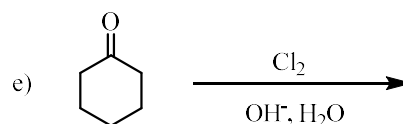
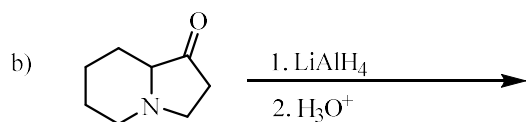
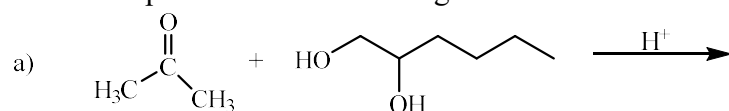
3. Predict the product of the following reaction.



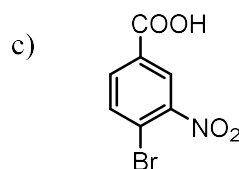
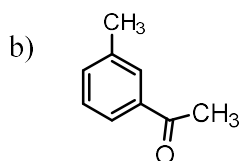
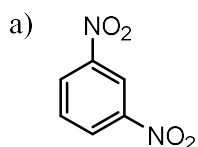
4. Give the structure of compounds A through H in the following series of reactions.



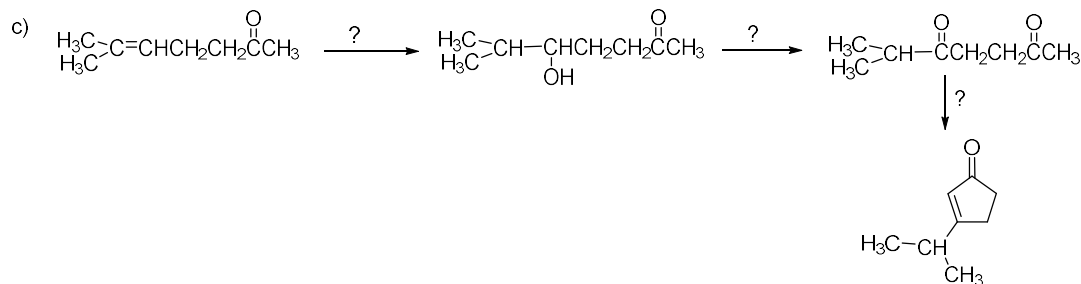
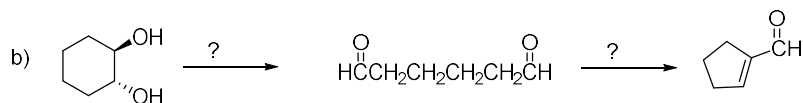
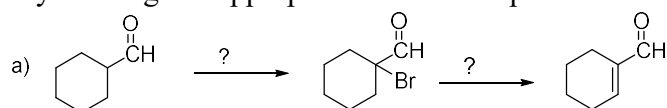
5. Predict the product of the following reaction.



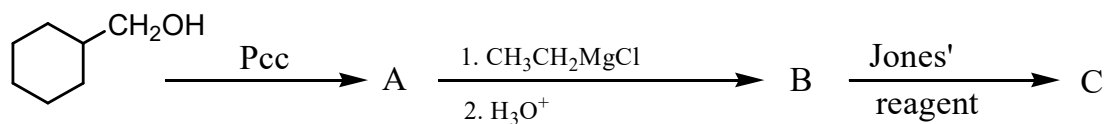
6. Derive synthesis of the following substituted benzene starting from benzene and using appropriate organic reagents.



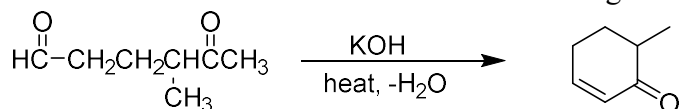
7. Identify the reagents appropriate for each step in the following syntheses:



8. Suggest structures for the missed products, reactants and or reagents of each reaction in the synthetic sequences.



9. Outline reasonable mechanisms for the following reaction.



10. Vitamin B₆ reacts with an enzyme (partial structure shown below) to form a coenzyme. Draw the structure of the coenzyme.

